2.3.5 $\forall n, q \in \mathbb{N}(q>0 \to \exists m, r: 0 \leq r < q \wedge n = mq + r)$
(Proposition 2.3.9)

q fixed. By the trichotomy of order, either $n<q$, $n=q$, or $n>q$

If $n<q$, $m=0$ and $r=n$ since any $m>0$ would make $mq+r>n$

If $n=q$, $m=1$ and $r=0$ since $1 \cdot q + 0 = n$ if $q=n$

If $n>q$, induct starting on $n=q+1$, implying $m=1$ and $r=1$
Assuming it is defined for n, now for $n++$, the equality holds by following this algorithm using the previous m and r:
- increment $r \to n++ = mq + r++$
  - if $r++ < q$, done
- if not, $r++ = q$, so m is incremented and $r=0$ since $n++ = mq + q = (m++)q$ by multiplication's definition